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Project Name: BoilerCompass

Problem Statement: Something that every motivated college student values is having a plan. In many departments at Purdue, students are provided a simple spreadsheet in which to manually form a plan of study for their time as an undergraduate. Even though myPurduePlan provides a platform for students to see courses they have taken and the requirements for graduation, there exists no easy way to visualize and validate a comprehensive plan of study that spans the duration of their college career. We will create a schedule planner that allows undergraduate students to effortlessly view and edit their degree plans, providing instant access to verified information such as prerequisites and credit hours. Students would be able to create groups to share their schedules and check for common classes. Further, schedule owners can edit access permissions to authorize other users to copy the schedules as templates.

Background:

- *Target Audience*:

  Our platform is designed for Purdue undergraduate students and advisors. Currently, undergraduate students have to discuss with their advisors before requesting courses on UniTime. Altering the schedule to find the best option can be very tedious, our tool can make degree and schedule planning easier and more intuitive.
- *Problem and Limitations of Current Similar Solutions*:

  Purdue undergraduates utilize myPurduePlan to have overviews of the requirements for their degrees, all courses taken, and what-ifs to evaluate what requirements a certain course fulfills. While myPurduePlan merely focuses on progress in the entire degree, other scheduling platforms, such as UniTime Scheduling Assistant and Coursicle allow students to check for credits and time of the courses, but only focus on planning the next semester. Furthermore, as Coursicle aims to integrate with hundreds of different universities, it offers no schedule or degree validation functionality for any of them. Thus, students currently must create their plans with spreadsheets or similar tools, manually researching course details and checking correctness. Advisors also have similar issues, where they would know and give advice to students' degree plans semester by semester only if students actually have made the plans.
- *Our Solution*:

  Instead of only providing insight into coursework already completed, or simply acting as a calendar application, BoilerCompass should be a tool to allow students to plan the future for their degree with confidence. The application will aggregate facts about Purdue courses from various sources in addition to the same

information regarding degree and credit requirements that myPurduePlan gives, and present this information to the user in the same platform used to plan their degree, from their upcoming semester to their final semester before graduation. Additionally, advisors will be able to create templates on BoilerCompass with the courses they commonly recommend to the students in the department they work in so that they can work more efficiently. Similarly, students will be able to seamlessly communicate their semester plans to their advisor by utilizing the same platform they use.

Requirements

Functional:

- **As a user (both students and advisors)**
    - I would like to create a personalized account with an email and password the first time I use the website so that I can get access to the service. This account creation includes verifying my email address.
    - I would like to log into my account with a Purdue email and password.
    - I would like to be able to reset my password if I forget it.
    - I would like to change my password on the settings page.
    - I would like to switch between dark/light modes on the settings page.
    - I would like to change my profile picture on my profile page.
    - I would like to have the option to delete my account whenever I choose to.
    - I would like to be able to save a schedule (either template or others' schedules that you have permission to see) for later reference.
    - I would like to see a descriptive graph that clearly delineates the order of the courses that should be taken.
    - I would like to customize the font, font size, and background.
    - I would like to enable two-step authentication for improved account security.
    - I would like to block and unblock other users.

- **As a student:**
    - I would like to begin using BoilerCompass without having to create an account yet, so that I can easily begin using the app.
    - I would like to create an account if I haven't already, so that I can access and edit my plans when I log in on other devices I own.
    - I would like to be able to easily export and import the entirety of my user data, so that I may migrate to a different device if I do not want to use BoilerCompass with an account.
    - I would like to access the profile page so that I can see details of my profile, including my name, profile picture, and my advisor's name. I would also like to be

able to choose my major, minor, degree focus, and expected graduation date. I will also be able to log out from the profile page.
- I would like to manually check which requirements I have completed, and which one I will complete when choosing future courses.
- I would like to see a summary of each semester, including the total credit hours, as well as total credit hours throughout the whole college life.
- I would like to upload my academic history from myPurdue Plan and have my degree progress automatically populated. (If time permits and conditions allow)
- I would like to have access to a course's information such as credits, prerequisites, instructor, and location.
- I would like to see syllabuses from the past for courses that I am considering taking. (If time permits and conditions allow)
- I would be able to send a request to an advisor, to give them permission and access to my schedule.
- I would like to create a group and share my schedule with other members. I would like to remove members from the group.
- I would like to be notified when I am adding a fall-through class to my schedule.
- I would like to be able to see a list of courses (under each category) and choose the ones I want to add to the specific semester on my schedule.
- I would like to be able to click on a specific course to see its information, including credit hours, course descriptions, and prerequisites.
- I would like to be able to search for the courses I want to add.
- I would like to be able to set a filter for courses and see the resulting courses based on my filters.
- I would like to be able to have the capability to easily add new courses to an already created schedule by simply dragging and dropping these courses into it.
- I would like to get a warning when I try to add a course for which I haven't completed the prerequisites, thereby getting denied to add that course.
- I would like to get a warning if my schedule has classes that have conflicting times.
- I would like to get a warning if my schedule exceeds the maximum credits permitted and turn off the warning after I get the permission.
- I would like to get a warning if the total credit hours of my schedule are less than the requirement to be a full-time student.
- I would like to download my plan in formats such as PDF or CSV.
- As a user, I would like to seamlessly/effortlessly transfer my course plans from my account into the UniTime course request system with a single action so that I only need to plan it once and have it in my request form.
- I would like to choose a template created by the advisors that contains all the suggested courses and timelines, and they can be populated into my schedule.

    - I would like to see data about grades achieved by students during past offerings of a particular course via Boiler Grades or university-provided public records request (ARPA), so that I can make an informed decision about the effect a course I take might have on my GPA.
    - I would like to know if a class is waivable with placement exams.
    - I would like to view all sessions of lectures and labs for a specific class in a certain semester.
    - I would like to have a separate page for a specific semester, in which I can add the sessions I want and generate a weekly schedule.
    - When creating a plan for a specific semester but course sessions have not been chosen, I would like to have auto-populated schedules that pick sessions based on starting time, ending time, or the most compact arrangement.
    - I would like to create and delete customized blocks that I can add to my weekly schedule.
    - For each block I created, I would like to change the name, duration, and color of the block.
    - I would like to be able to see a recommended schedule for my major, and the schedule will be able to be rearranged as I put in my transfer credits.
    - I would like to experiment with changing/adding different majors and see the path of graduation to help me visualize this hypothetical idea.
    - I would like to be able to tag/mark classes that I am not sure if I want to take or not and access them later.
    - I would like to be able to enter my grades for the courses I already took and the expected (or goal) grades for future courses and see the resulting GPA.

- **As an advisor:**
    - I would like to access the profile page so that I can see details of my profile, including my name, profile picture, and department. I will also be able to log out from the profile page.
    - I would like to see a list of my students, and click on each of the students to see more information about each student. The information should be as much as the students can see (aside from personal settings)
    - As I look at the students' information and their plans, I would like to make suggestions and comments on their schedule. The students can choose to accept them or not. This is useful when students meet with their advisors, and they can accept the changes.
    - I would like to send requests to students, so they can permit me to access their schedule.
    - I would like to create a planning template for different majors prepopulated with suggested courses and timelines for students to use.

- I would like to compare the schedules of all my current students to give better suggestions.
- I would like to save the schedules of all my past students to understand how students' preferences may have changed.
- I would like to track and monitor students' progress toward degree completion.
- I would like to conduct virtual advising sessions through video conferences or chat functionalities when I want to have a quick talk with students.

- Non-functional:
  - Website capacity:
    - The website should be able to function properly under the capacity of 50,000 Purdue students and advisors using it at the same time.

  - Data privacy:
    - The website should be able to protect all students' personal schedule data from being accessed by outsiders without the students' permission. This will be done via encryption and authentication/privacy controls using Firebase.

    - The database of the website should be safely protected and prevent unauthorized access to users' information.

    - The verification process for re-setting passwords and verifying passwords should be secure and reliable.

  - Running and loading time:
    - The website should be able to locate and present a student's schedule under 5 seconds.

  - Accessibility: All Purdue students and advisors verified by their Purdue email should have access to the website 24 hours a day.